

ME

Mohamed Ezer Bouzouraa

Software & AI Engineer

CONTACT

Location
Tunis, Tunisia

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LANGUAGES

ArabicNative
FrenchFluent
EnglishHigh Level
ItalianIntermediate

Mohamed Ezer Bouzouraa

Software & AI Engineer

Self-taught engineer specializing in AI/ML — designing, training & deploying intelligent data-driven solutions.

EDUCATION

Engineering - Software & AI Specialization2024 - Present
National Institute of Applied Sciences and Technology (INSAT)
Engineering Preparatory Cycle - Mathematics, Physics & Informatics
Specialization: Machine Learning & Deep Learning

Baccalaureate - Tunisian Curriculum2024
Lycée Salah Ben Youssef
Graduated with distinction — score: 18.28 / 20

PROJECTS

— Agentic AI & LLMs

INSAT RAG Assistant [LangChain · BM25 · Groq]
Multilingual chatbot using RAG with BM25 re-ranking for INSAT-specific knowledge.
RLHF World Cup 2026 [Llama 3.3 · Groq · RLHF]
World Cup chatbot trained with RLHF feedback for continuously improving response quality.
Reviewify [LLM Agent]
Agentic pipeline converting bullet-point notes into polished employee performance reviews.
Student Advisor AI [LLM · UX]
Step-by-step learning companion adapting explanations to each student's pace.
MatchPicker AI Dashboard [Flask · AI Chatbot]
Football analytics dashboard with live match data and contextual AI-powered analysis.
Ilama-chatbot [Gradio · TinyLlama-1.1B]
Sleek Gradio-based chatbot running TinyLlama-1.1B locally or on GPU, with a clean dark-mode UI and a modular, easily extensible structure.

— Computer Vision

YOLOv8 Car Counter [YOLOv8 · OpenCV]
Real-time vehicle detection and counting with bounding-box visualization.
Push-Up Counter [MediaPipe · Adaline]
Pose estimation pipeline measuring joint angles with Adaline to count push-ups.
Soccer Team Detection [YOLOv8 · Color Classification]
Classifies soccer players into teams via jersey-color analysis using YOLOv8.
SmartCribGuardian [YOLOv9 · OpenCV]
Real-time baby crib monitoring system that detects when a baby leaves the crib, drawing bounding boxes and sending alerts; includes webcam capture and configurable crib zones.

— Classical Machine Learning

Lifestyle Stability Index [SVC · GridSearchCV]
SVM classifier with hyperparameter tuning to categorize daily routines as stable/unstable.
Geo Country Guesser [Decision Tree · Tkinter]
Desktop game inferring a country from 15 geographic/cultural features.
SpamGuard [SVM · TF-IDF]
Spam detection app using TF-IDF vectorization and a kernel-SVM classifier.
Vehicle Health Predictor [Logistic Regression]
End-to-end ML pipeline classifying vehicle health from diagnostic sensor data.

— Full-Stack & Web

Mathematicians Legacy [Symfony · Flask · Gemini AI]
Interactive world map of mathematical geniuses with a Gemini AI chatbot narrating contributions.
All projects available at github.com/mohamedezerbouzouraa

BLOGS

How Mathematicians Know Something Is True Forever [Philosophy of Mathematics · Essay]
Explores the nature of mathematical proof — why a proven theorem stays true eternally — covering proof strategies, Gödel's incompleteness, and the aesthetics of elegant proofs.
The Infinite Mystery of Prime Numbers [Prime Numbers · Number Theory · History of Ideas]
From Fermat's beautiful conjecture — shattered by Euler — to Wilson's elegant divisibility criterion and the deepest unsolved problems in mathematics.

TECHNICAL SKILLS

AI / ML
TensorFlow · PyTorch · scikit-learn · LLMs · RAG · RLHF · Deep Learning
Languages
Python · JavaScript · C/C++ · Java · PHP · SQL
Data Science
NumPy · Pandas · Matplotlib · MediaPipe · YOLOv8
Engineering
Flask · Symfony · LangChain · Git · Linux · REST APIs

ACHIEVEMENTS

Created and ran a technology-focused YouTube channel (2017), growing it to roughly 4,000 subscribers (channel is currently deleted).